DESIGN AND FABRICATION OF AN AUTOMATED CHAIR CUM BED MECHANISM FOR BEDRIDDEN PATIENTS

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

in

Mechanical Engineering

of

APJ Abdul Kalam Technological University

by

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April 2026

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Certificate

This is to certify that the Project Report titled **"DESIGN AND FABRICATION OF AN AUTOMATED CHAIR CUM BED MECHANISM FOR BEDRIDDEN PATIENTS"** is a bonafide record of the work carried out by **ABIN C S (VAS22ME006), BLESON JOSEPH (VAS22ME024)** and **GODSON V O (VAS22ME026)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology** in **Mechanical Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2025-2026. The project report has been approved as it satisfies the academic requirements in the respect of project work prescribed for the said degree.

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We, **ABIN C S (VAS22ME006), BLESON JOSEPH (VAS22ME024)** and **GODSON V O (VAS22ME026)**, hereby undertake that the project work entitled **"DESIGN AND FABRICATION OF AN AUTOMATED CHAIR CUM BED MECHANISM FOR BEDRIDDEN PATIENTS"** is carried out by us independently under the valuable guidance of **Mr. VISHNU SHAJI**, Asst. Prof., MED and **Dr. RENJISH VIJAY**, Assoc. Prof., MED of Vidya Academy of Science and Technology, Thalakottukara, Thrissur, in partial fulfillment of the requirements for the award of degree of **Bachelor of Technology in Mechanical Engineering of APJ Abdul Kalam Technological University**, during the academic year 2025-2026.

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April 2026

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During the course of our main project work, several people collaborated directly and indirectly with us. Without their support it would be impossible for us to finish our work. That is why we wish to dedicate this section to recognize their support.

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Abstract

This project details the design, simulation and fabrication of an automated chair cum bed mechanism engineered to enhance the autonomy and comfort of bedridden patients. Addressing the inherent limitations of conventional hydraulic actuated systems specifically fluid leakage and high maintenance requirements this study implements a robust rack and pinion transmission driven by a high torque electric motor. The mechanical assembly was developed using 3D CAD modeling and validated through structural analysis to ensure load bearing capacity and ergonomic stability. Operation is facilitated by an Arduino based microcontroller integrated with a voice recognition module, providing a hands free, intuitive interface for patients with limited mobility. Safety protocols are enforced through the integration of limit switches for overtravel protection and a manual override system for emergency use. Comparative evaluation demonstrates that this electromechanical approach offers a more durable, cost effective and hygienic alternative to existing medical furniture, successfully validating the feasibility of an integrated, automated solution for homecare and clinical environments.

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List of Symbols and Abbreviations

M	Mass of patient (kg)
g	Acceleration due to gravity (m/s^2)
W	Total static load (N)
FoS	Factor of Safety
F_d	Design load (N)
F_{back}	Load on backrest (N)
F_t	Tangential force on pinion (N)
m(module)	Gear module (mm)
Z	Number of teeth (pinion)
d	Pitch circle diameter (mm)
b	Face width (mm)
σ_b	Allowable bending stress (MPa)
y	Lewis form factor
T	Torque / torsional moment (Nmm)
M_b	Bending moment (Nmm)
M_e	Equivalent twisting moment (Nmm)
$\tau(tau)$	Allowable shear stress (MPa)
CAD	Computer Aided Design
DC	Direct Current
M.S	Mild Steel

Chapter 1

INTRODUCTION

1.1 Background

Hydraulic systems in hospital beds are heavy, complex, and prone to leakage. They require frequent maintenance and offer limited precision in motion control, especially during posture adjustments. The presence of hydraulic fluid also raises hygiene concerns in clinical and home care environments, as leaks can contaminate bedding and pose safety risks. Moreover, manual or valve-based hydraulic controls often lack ergonomic accessibility for patients, requiring caregiver intervention for even minor adjustments.

This project replaces hydraulic actuation with a rack-and-pinion mechanism driven by an electric motor, offering a compact, fluid-free alternative. The rack-and-pinion setup converts rotary motion into linear displacement, enabling smooth and controlled movement of the bed's head and leg sections. Coupled with a worm gearbox, the system achieves high torque output with self-locking capability, ensuring stability even when power is off. The electromechanical design reduces maintenance, improves hygiene, and allows integration with voice-controlled automation for enhanced user independence.

1.2 Motivation

To create a compact, fluid-free, low-maintenance chair-cum-bed system that improves hygiene, safety, and user comfort, especially for bedridden and mobility-impaired individuals. Traditional hospital beds often rely on hydraulic systems that are bulky, leak-prone, and require frequent maintenance, which can compromise patient safety and in-

crease operational costs. These systems also demand manual effort or valve-based control, limiting precision and independence for users.

By adopting a rack-and-pinion mechanism driven by an electric motor, this project aims to eliminate fluid-based actuation entirely, reducing the risk of contamination and simplifying servicing. The compact mechanical layout allows for space-efficient installation in both hospital and home environments. Integration of voice control further enhances accessibility, enabling patients to adjust their posture without physical strain or caregiver intervention. The system prioritizes ergonomic comfort, modularity, and affordability, making it a practical solution for inclusive healthcare design.

1.3 Mechanism Overview

- **Rack-and-pinion converts rotary motion to linear displacement:** The pinion gear, driven by the motor, engages with the rack to produce smooth linear motion. This setup allows precise control over the movement of bed sections, making it ideal for posture adjustment in assistive devices.

- **Electric motor ensures precise control:** A 12V DC motor is used to drive the pinion, offering consistent torque and speed regulation. When paired with a worm gearbox, the motor provides self-locking capability and enables accurate positioning without drift.

- **Eliminates hydraulic fluid, reducing mess and maintenance:** By removing the need for hydraulic cylinders and fluid reservoirs, the system avoids leakage risks and simplifies cleaning. This improves hygiene in patient environments and reduces long-term servicing costs.

- **Worm gear provides high torque and self-locking:** The integration of a worm gearbox not only multiplies torque but also prevents back-driving, ensuring that the bed remains securely in position even when power is off.

- **Compact and lightweight design:** The rack-and-pinion system requires minimal space compared to hydraulic setups, allowing for a more compact and lightweight mechanism suitable for portable assistive furniture.

- **Smooth and silent operation:** The gear engagement is optimized for minimal vibration and noise, ensuring user comfort, especially in hospital or home-care environments.
- **Simple electronic control:** The direction and speed of movement can be easily controlled using a simple switch or a microcontroller-based system, enabling automated adjustment functions.
- **Cost-effective and energy efficient:** Electric actuation consumes less power than hydraulic pumps and does not require frequent refilling or maintenance, reducing overall operational costs.
- **Safety and reliability:** The self-locking property of the gearbox and mechanical limit switches ensure the bed stops safely at predefined positions, preventing over-extension and mechanical strain.
- **Modular structure for easy assembly:** Each mechanism unit (motor, rack, pinion, and gearbox) can be independently assembled or replaced, simplifying maintenance and enhancing system flexibility.



Figure 1.1: Rack-and-Pinion Mechanism for Chair-Cum-Bed

Chapter 2

LITERATURE REVIEW

2.1 Background

The evolution of patient care has been defined by the pursuit of improving the quality of life for individuals with restricted mobility. Traditionally, caring for bedridden patients—those with paralysis or spinal cord injuries—relied on manual intervention. Caregivers must physically lift patients to change posture, a labor-intensive process posing musculoskeletal risks to both parties.

In clinical settings, preventing pressure ulcers and respiratory issues requires frequent positioning changes. While standard hospital beds offer elevation, transitioning between lying and sitting typically requires transferring the patient to a separate chair. This transfer is a primary source of physical strain. Current hydraulic and manual beds are often hindered by high maintenance, fluid leakage, and a lack of user autonomy.

The ”Smart Healthcare” paradigm now integrates microcontrollers into assistive devices. Using Arduino and robust mechatronics, there is an opportunity to develop ”active” furniture. An automated chair-cum-bed mechanism addresses the dual challenge of mobility and comfort. Replacing hydraulics with rack-and-pinion systems and voice interfaces reduces dependency, enhances patient dignity, and optimizes operational efficiency in home-care and hospital environments.

2.2 Review of Literature

Research on voice-controlled assistive mobility devices has gained significant attention in recent years. A study by D. Vijendra Babu, P. Subramanian, and R. Karthikeyan (2018) in the International Journal of Pure and Applied Mathematics presented a voice-based control system for a wheelchair using speech recognition. The system employed a Bluetooth module to transmit voice commands to the control unit and utilized Hidden Markov Models to achieve accurate command recognition, enabling independent movement for disabled and elderly users.

Another study by K. Velusamy et al. (2019) in the International Journal of Engineering Science and Computing (IJESC) developed a voice-controlled wheelchair using an Arduino microcontroller and speech recognition module. The design incorporated ultrasonic sensors for obstacle detection and avoidance. The system could be operated through both graphical and voice commands via a smartphone interface, demonstrating an affordable and practical assistive technology solution.

Tan Kian Hou, Yagasena, and Chelladurai (2020) presented an Arduino-based voice-controlled wheelchair in the Journal of Physics: Conference Series. Their design converted a manual wheelchair into an automated prototype using a voice recognition module that translated speech signals into digital commands. The inclusion of Bluetooth communication and optional joystick control improved flexibility and user convenience.

In another study, M. D. Balsaraf, V. S. Takate, and Bachkar Siddhant (2017) proposed an Android-based wheelchair control system in the International Journal of Advances in Scientific Research and Engineering (IJASRE). The system used a smartphone interface connected through an IOIO board to control DC motors, allowing users to navigate the wheelchair through mobile commands.

Similarly, Prof. Manoj V. Bramhe et al. (2017) in the International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering (IJAREEIE) developed a voice-controlled wheelchair specifically designed for severely disabled individuals. The system used an HM2007 voice recognition module connected to an Arduino controller, enabling patients with limited physical mobility to

control movement using voice commands.

These studies highlight the growing role of voice recognition and microcontroller-based systems in assistive technologies for people with mobility impairments. However, most of the existing research focuses primarily on wheelchair mobility. There is relatively limited work addressing automated mechanisms that assist bedridden patients in changing posture or transitioning between bed and chair configurations. Therefore, the present project aims to extend the concept of voice-controlled assistive technology by developing an automated chair–cum–bed mechanism using a rack-and-pinion drive system combined with a voice-controlled interface.

2.3 Findings from Literature and Gap Analysis

The review of existing literature establishes that significant progress has been made in using Arduino microcontrollers and voice recognition modules to automate mobility aids, particularly wheelchairs. Research has consistently shown that low-cost components like DC motors, Bluetooth modules (HC-05), and ultrasonic sensors can effectively replace traditional manual systems to provide independence to disabled users. However, a notable gap exists in the literature regarding the application of these voice-controlled systems to multi-functional furniture, specifically for bedridden patients who require more than just horizontal or vertical movement. While wheelchairs address transportation, there is limited research on automated mechanisms that can seamlessly transition between a ”chair” and a ”bed” configuration to alleviate pressure sores and improve patient posture. Most existing hospital beds rely on complex, high-maintenance hydraulic systems that are prone to leakage. This project addresses this gap by introducing a voice-controlled, motor-driven rack-and-pinion mechanism with a worm gearbox, providing a more reliable, precise, and hygienic solution for long-term bedridden care.

Chapter 3

OBJECTIVES AND SCOPE

3.1 Objectives

The primary objective of this project is to design and fabricate a voice-controlled, automated chair-cum-bed mechanism that provides bedridden patients with an independent and reliable means of adjusting their posture.

The specific technical objectives are:

- **To design a dual-configuration structural framework:** To develop a robust mechanical assembly capable of transitioning seamlessly between a standard hospital bed and a seated chair configuration while maintaining ergonomic stability for the patient.
- **To implement a high-torque mechanical drive system:** To replace traditional hydraulic actuators with a rack-and-pinion mechanism powered by a worm gearbox, ensuring precise motion control and reducing maintenance requirements.
- **To integrate an Arduino-based voice control interface:** To develop a microcontroller-based system that processes oral commands via a voice recognition module, allowing for hands-free operation and enhanced user autonomy.
- **To incorporate safety and feedback protocols:** To install limit switches, ultrasonic sensors, and manual override features to prevent mechanical over-travel and ensure patient safety during the configuration change.

- **To evaluate performance and cost-effectiveness:** To conduct structural and ergonomic analyses to verify the system's load-bearing capacity and energy efficiency, ensuring the final prototype is a viable, low-cost alternative for home-care and clinical environments.

3.2 Problem Definition

The care of bedridden patients remains a significant challenge in both clinical and home-care settings due to the inherent limitations of traditional medical furniture. The primary issues addressed by this project are:

- **Dependency and Patient Discomfort:** Bedridden patients with restricted mobility are entirely dependent on caregivers for basic postural adjustments. This lack of autonomy often leads to psychological distress and physical discomfort, as patients cannot independently change from a lying to a sitting position.
- **Operational Inefficiency of Current Systems:** Existing adjustable hospital beds typically utilize hydraulic or manual crank mechanisms. Hydraulic systems are prone to fluid leakage, high maintenance costs, and mechanical complexity, while manual systems require significant physical exertion from caregivers, leading to potential musculoskeletal injuries.
- **Health Risks of Immobility:** Prolonged static positioning increases the risk of pressure ulcers (bedsores) and respiratory complications. The difficulty of transitioning a patient from a bed to a separate chair often results in infrequent movement, exacerbating these health risks.
- **Interface Limitations:** Standard motorized beds often use handheld pendants or joysticks which may be difficult for patients with limited hand dexterity or upper-limb disabilities to operate. There is a clear need for a hands-free, intuitive interface—such as voice control—to bridge this accessibility gap.
- **Cost and Complexity:** Many advanced automated medical beds are prohibitively expensive for home-care environments. There is a lack of cost-

effective, reliable solutions that combine automation, structural stability, and ease of use in a compact, hygienic design.

In contrast, the current healthcare infrastructure lacks an integrated, low-maintenance, and voice-activated mechanism that allows a single piece of furniture to serve as both a bed and a chair, providing independence to the patient while reducing the burden on the caregiver.

3.3 Scope and Limitations

The care of bedridden patients is currently hindered by the limitations of traditional medical furniture. The primary challenges addressed by this project include:

- **Loss of Autonomy:** Patients with restricted mobility are entirely dependent on caregivers for basic postural adjustments, leading to physical discomfort and psychological distress.
- **Inefficiency of Current Systems:** Existing hospital beds primarily use manual cranks or hydraulic systems. Manual beds cause caregiver strain and injury, while hydraulics are prone to fluid leakage and high maintenance costs.
- **Health Risks:** Prolonged static positioning increases the risk of pressure ulcers (bedsores) and respiratory complications. Frequent transitioning between bed and chair is often avoided due to the physical difficulty of the transfer.
- **Interface Barriers:** Motorized beds often use hand-held joysticks or buttons, which are inaccessible to patients with limited hand dexterity.
- **Cost and Complexity:** Advanced automated solutions are often too expensive for home-care or small clinics, creating a need for a cost-effective, multi-functional alternative.

In Contrast, there is a critical need for an integrated, voice-activated mechanism that transitions between bed and chair configurations to provide patient independence and reduce caregiver burden.

Chapter 4

DESIGN AND DRAWING

4.1 Load Analysis and Design constant

Patient mass (m) = 100kg

Acceleration due to gravity (g) = 9.81m/s^2

$$\begin{aligned} Total Static weight(W) &= m \times g \\ &= 9.81N \end{aligned}$$

Factor of Safety(Fos) = 2.0 (to account for dynamic patient movement)

$$\begin{aligned} Design Load(F_d) &= W \times Fos \\ &= 981 \times 2.0 \\ &= 1962N \end{aligned}$$

$$\begin{aligned} BackrestLoad(F_{back}) &= 58\%ofW \\ &= 0.58 \times 981 \\ &= 568.98 \end{aligned}$$

(Based on torso weight distribution) = 569 N

4.2 Design of Pinion

Material = Cast Iron

Module(m) = 1.5mm

Number of Teeth(Z) = 21

$$\begin{aligned} PitchcircleDiameter(d) &= m \times Z \\ &= 31.5mm \end{aligned}$$

$$\begin{aligned} OuterDiameter(d_o) &= d + 2m \\ &= 31.5 + 2 \times 1.5 \\ &= 34.5mm \end{aligned}$$

$$\begin{aligned} Totaldepth(h) &= 2.25 \times m \\ &= 2.25 \times 1.5 \\ &= 34.5mm \end{aligned}$$

$$\begin{aligned} Tangential Force(F_t) &= Assuming two racks \\ F_t &= F_{back}/2 \\ &= 284.5N \end{aligned}$$

4.2.1 Lewis Strength Verification

$$w_t = \sigma_b \times b \times \pi \times m \times y \tag{4.1}$$

Allowable Stress (σ_b)= $60MP_a$ (Cast Iron)

Face Width (b) = $15mm(10 \times m = 10 \times 1.5)$

Form Factor (y) = 0.32 (For Z=21)

$$\begin{aligned} W_t &= 60 \times 15 \times \pi \times 1.5 \times 0.32 \\ &= 60 \times 15 \times 4.71 \times 0.32 \\ &= 900 \times 1.5072 \\ &= 1356.48N \end{aligned}$$

Since the allowable strength 1356.48N is significantely higher than the actual applied load 284.5N, the cast iron pinion with 1.5 module and 21 teeth is safe for a 100kg patient

As Calculated earlier,

$$\begin{aligned} Total force of patient &= 100kg \times 9.81 \\ &= 981N \end{aligned}$$

Force on Backrest (58% of weight) = 569N

$$\begin{aligned} Force per Pinion(F_t) &= 569/2 \\ &= 284.5N \end{aligned}$$

The actual tangential for per pinion for a 100kg patient is 284.5N

$$\begin{aligned} SafetyFactorcheck &= \frac{1356.48}{284.5} \\ &= 4.7 \end{aligned}$$

The design is safe as the permissible load exceeds the actual load by a factor of 4.7

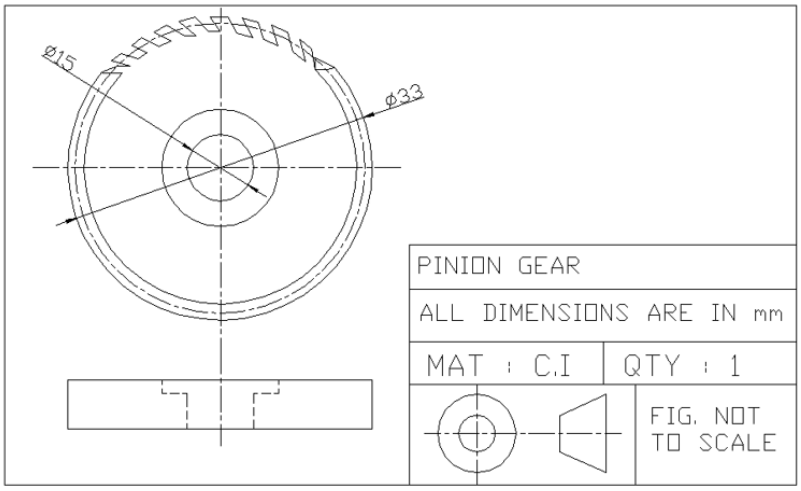


Figure 4.1: 2D drawing of Pinion

4.3 Design of Rack

Module(m) = 1.5mm

cross-section $= 20 \times 15mm$

Circumference or Rotation= $\pi \times d = 97.38mm$

Actual Length Provided= 215mm

Number of Teeth(Z_r)= 45

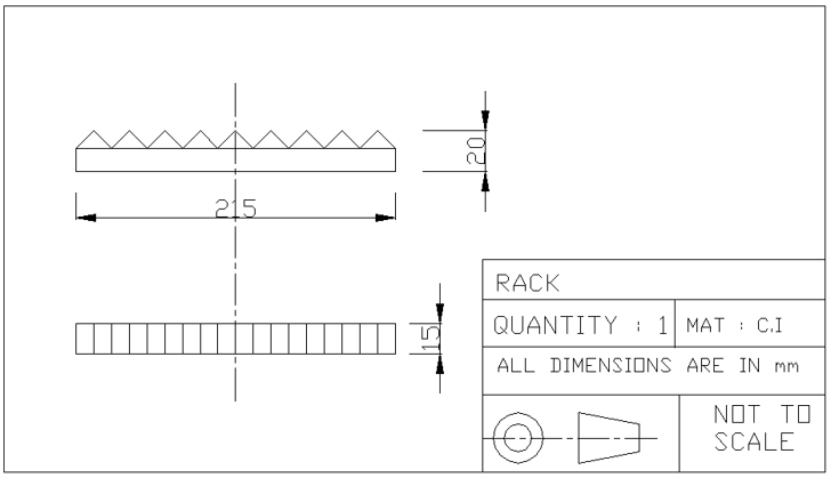


Figure 4.2: 2D drawing of Rack

4.4 Design of Shaft and Bearing

Patient mass(m)=100kg

$$\begin{aligned} DesignLoad(w) &= m \times g \\ &= 100 \times 9.81 \\ &= 981N \end{aligned}$$

Load Per Shaft= Assuming the mechanism uses two shafts or a split load, F= 490.5N

Motor Power(P)= $2HP \approx 1.5Kw$

Motor Speed(N)= 1440 rpm

Material Allowable Shear stress(τ)= $40MPa(40N/mm^2)$

4.4.1 Calculation of Tortional Moment(M_t)

$$\begin{aligned} T &= \frac{60 \times P}{2\pi \times N} \\ T &= \frac{60 \times 1500}{2\pi \times 1440} \\ &= 9.947 \approx 10Nm \end{aligned} \tag{4.2}$$

Note; (in Nmm for stress calculations, 10,000Nmm)

4.4.2 Calculation of Bending Moment(M_b)

$$\begin{aligned} M_b &= F \times L \\ M_b &= 490.5 \times 40mm \\ &= 19620Nmm \end{aligned} \tag{4.3}$$

4.4.3 Equivalent Twisting moment (M_e)

$$\begin{aligned} M_e &= \sqrt{(M_b)^2 + (T)^2} \\ &= \sqrt{(19620)^2 + (10,000)^2} \\ &= \sqrt{384,944,400 + 100,000,000} \\ &= \sqrt{484,944,400} \\ &= 22,021.5 \text{ Nmm} \end{aligned} \tag{4.4}$$

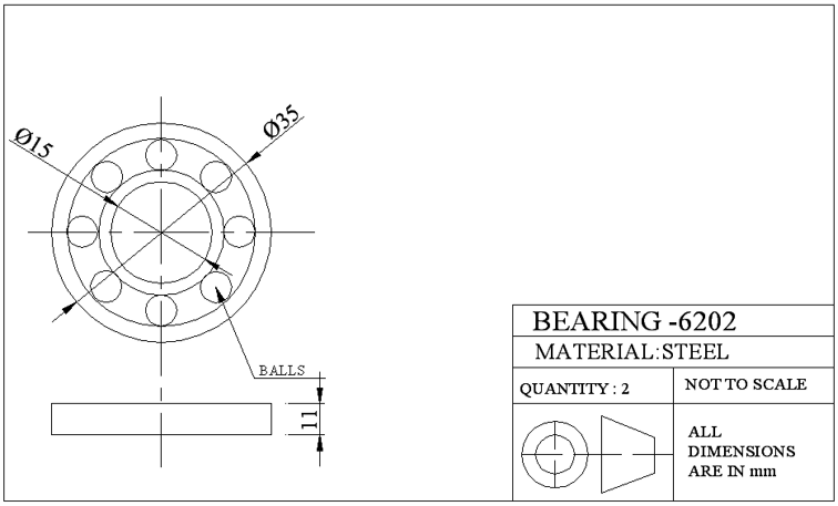


Figure 4.3: 2D drawing of Shaft

4.5 Determination of Minimum Shaft Diameter(d)

$$
\begin{aligned}
\tau &= \frac{16 \times M_e}{\pi \times d^3} \qquad (4.5)\\
d &= \sqrt[3]{\frac{16 \times M_e}{\pi \times \tau}}\\
&= \sqrt[3]{\frac{16 \times 22021.5}{\pi \times 40}}\\
&= \sqrt[3]{\frac{352344}{125.66}}\\
&= \sqrt[3]{2803.9}\\
&= 14.1\text{mm}
\end{aligned}
$$

Calculated Minimum Diameter = 14.1mm

Actual Shaft Diameter Selected = 15mm

Since the actual diameter is greater the calculated minimum required diameter (14.1mm), The shaft is safe and will not undergo permanent deformation or failure under the weight of a 100kg patient.

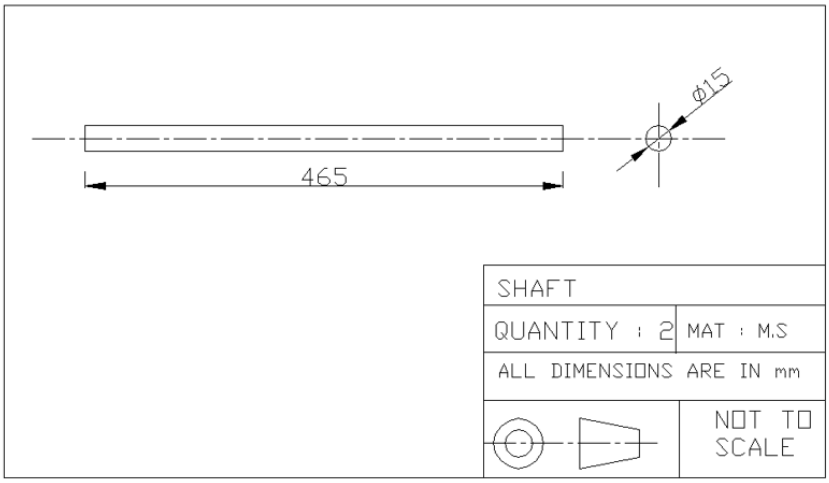
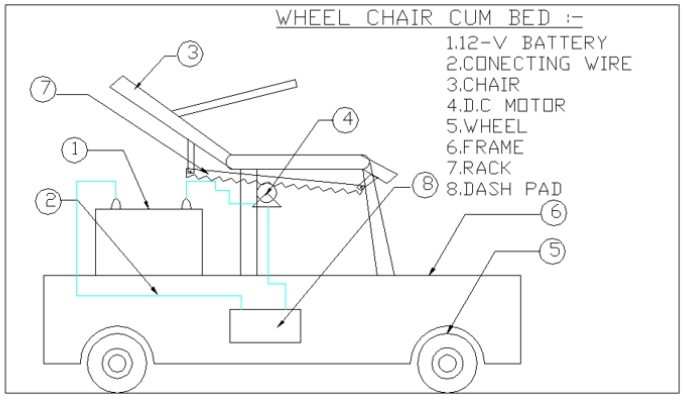
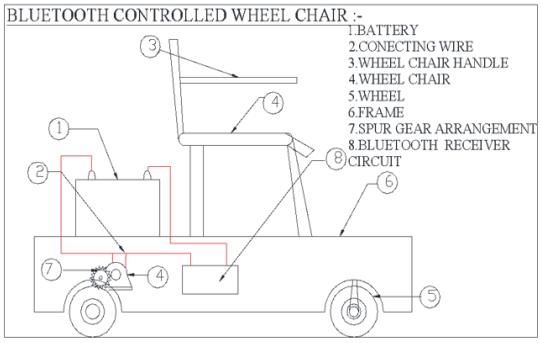


Figure 4.4: 2D drawing of Shaft



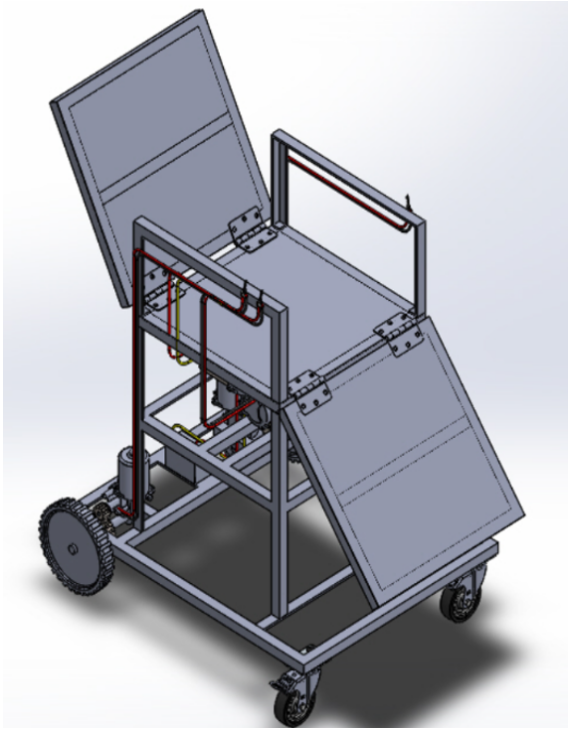
(a) 2D drawing of Wheelchair cum bed with rack and pinion mechanism



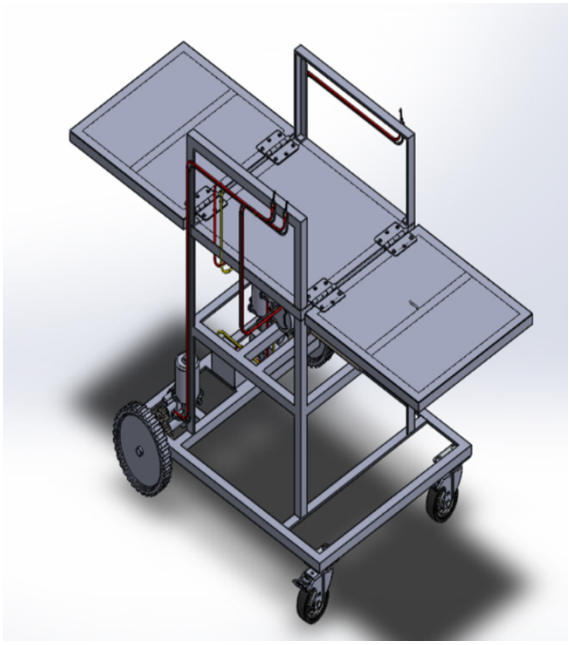
(b) 2D drawing of Wheelchair cum bed with shaft and gear mechanism

Figure 4.5: 2D drawing of Wheelchair cum bed

4.6 3D CAD Model and Assembly Views

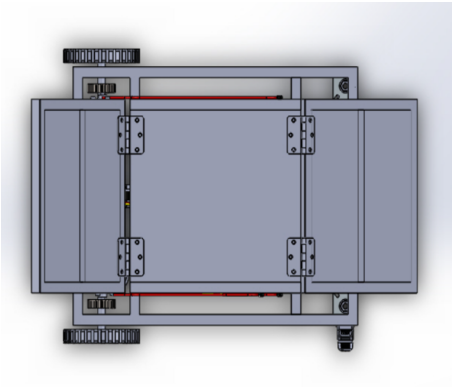


(a) Front view 3D model of wheelchair cum bed before stretching the bed

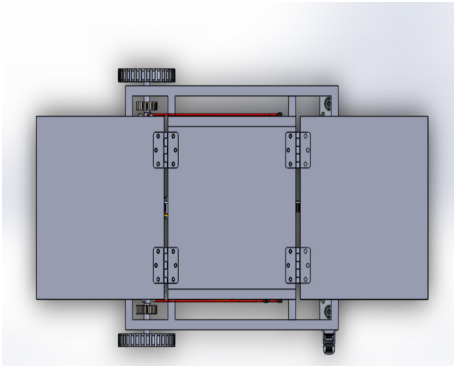


(b) Front view 3D model of wheelchair cum bed after stretching the bed

Figure 4.6: Front view of the 3D SolidWorks model showcasing the chair configuration.



(a) Seated (Chair) configuration



(b) Extended (Bed) configuration

Figure 4.7: Top view of the 3D SolidWorks model showcasing the chair configuration.

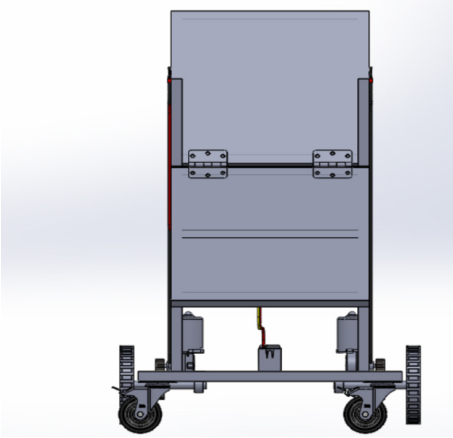
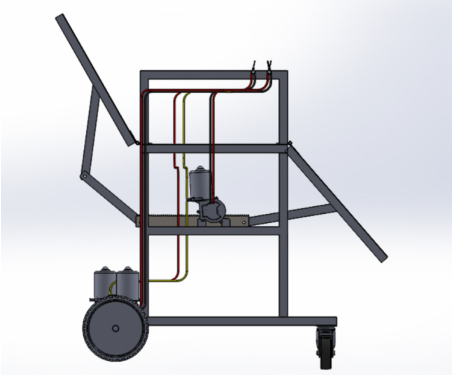
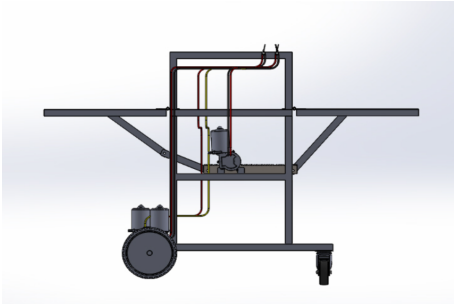


Figure 4.8: Rear view of the 3D CAD model, detailing the worm gearbox and motor mounting configuration.



(a) Before stretching the bed (Chair configuration)



(b) After stretching the bed (Bed configuration)

Figure 4.9: Side view analysis of the chair-cum-bed mechanism showing the range of motion.

Chapter 5

FABRICATION AND ASSEMBLY

5.1 Selection of Materials

The selection of materials for the proposed chair–bed mechanism was carried out based on structural strength, durability, patient safety, and the mechanical requirements of the rack-and-pinion drive system. The materials chosen for various components are described below.

Mild Steel (M.S.): Mild steel was used for the primary chassis and the chair–bed frame. This material was selected due to its high tensile strength, good weldability, and excellent load-bearing capacity. It is capable of supporting the weight of a bedridden patient without experiencing significant structural deformation.

Cast Iron: Cast iron was selected for the rack and pinion gears due to its high compressive strength and superior wear resistance. These properties make it suitable for gear applications where repeated contact and load transmission occur. The material also performs well under the calculated contact stress conditions.

$C45Steel$**:** $C45steel$ was used for the 15 mm diameter shaft. This material was selected because of its good mechanical strength and torsional rigidity, which enables it to transmit the required torque of $10Nm$ from the motor to the pinion without failure.

Lead-Acid Battery: A $12V, 7Ah$ lead-acid battery was chosen as the power source for the system. This battery provides a stable and reliable DC supply required to

operate the high-torque DC motor as well as the control electronics used in the automation system.

5.1.1 Bill of Materials (B.O.M)

Table 5.1: Bill of Materials

Sl. No.	Name of the Component	Material	Quantity
01	12V DC Motor	Aluminium	3
02	12 V Battery 7 AH	Lead-Acid	1
03	Shaft	M.S	2
04	Chassis	M.S	1
05	Chair Arrangement	Pipe & Channel (M.S)	1
06	Bolt, Nut and Washer	Steel	10
07	Rack and Pinion Gear	M.S	1
08	Wheel	Fiber	4
09	Bearing	Steel	4
10	Bearing Cap	M.S	4
11	Spur Gear	C.I	4

5.2 Manufacturing Processes

The fabrication of the prototype was carried out through a series of standard mechanical manufacturing processes. These processes ensured dimensional accuracy, structural stability, and proper functioning of the mechanism.

Marking and Measuring: All required dimensions for the frame, shaft (15 mm), and rack (215 mm) were carefully marked on the raw material based on the prepared 2D engineering drawings.

Cutting and Machining: The mild steel sections used for the frame were cut to the required dimensions using a power hacksaw. The shaft was then machined on a lathe to obtain the precise diameter required for proper fitting with the 6202 ball bearings.

Welding: Arc welding was used to join the structural members of the base frame and the adjustable backrest mechanism. This process ensured a rigid and permanent connection between the structural components.

Drilling: Drilling operations were performed to create accurate mounting holes for the motor assembly, bearing housings, and pivot joints required for the chair–bed transformation mechanism.

Finishing: After fabrication, all components were ground to remove sharp edges and burrs. The frame was then coated with an anti-corrosive primer followed by a protective paint layer to improve durability and maintain hygiene in a medical environment.

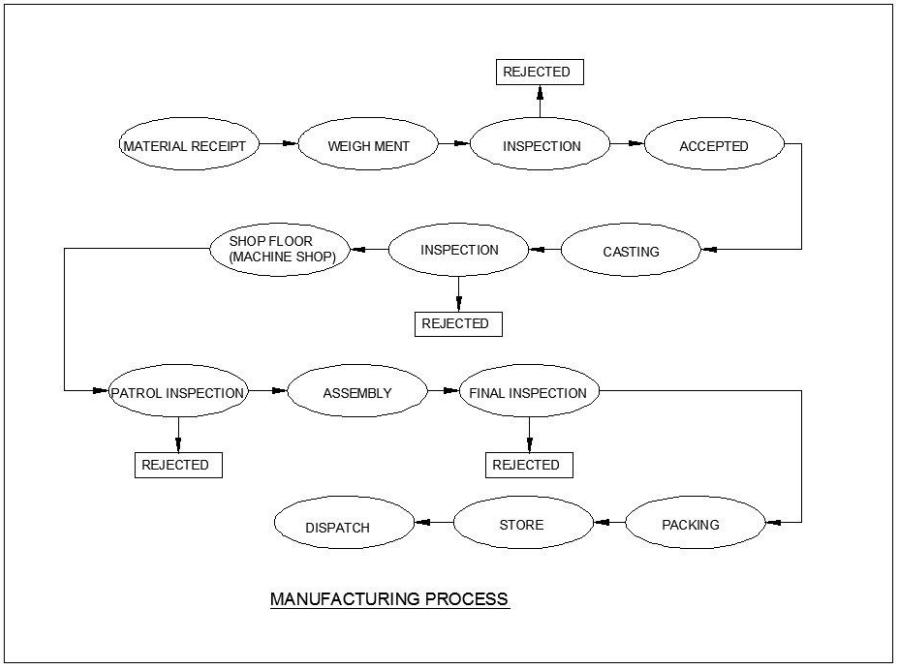


Figure 5.1: Manufacturing Processes

5.3 Electronic Integration

The automation of the chair–bed mechanism was achieved through the integration of an electromechanical control system.

Control Unit: An Arduino microcontroller was used as the central control unit of the system. It was programmed to receive input signals from the voice recognition module and to control the motor accordingly.

Actuation System: A high-torque DC motor equipped with a worm gearbox was used to drive the pinion gear. The worm gear mechanism provides a self-locking property, which prevents the bed from moving unintentionally when power is disconnected.

Voice Interface: The voice recognition module processes oral commands such as ”Bed” and ”Chair” and transmits the corresponding signals to the Arduino controller. Based on the received command, the controller activates the motor driver to rotate the pinion either clockwise or counterclockwise.

Safety Feedback System: Limit switches were installed at both ends of the rack travel path (approximately 194.76 mm). These switches automatically cut off the motor power when the mechanism reaches its maximum allowable displacement, thereby preventing mechanical over-travel and ensuring safe operation.

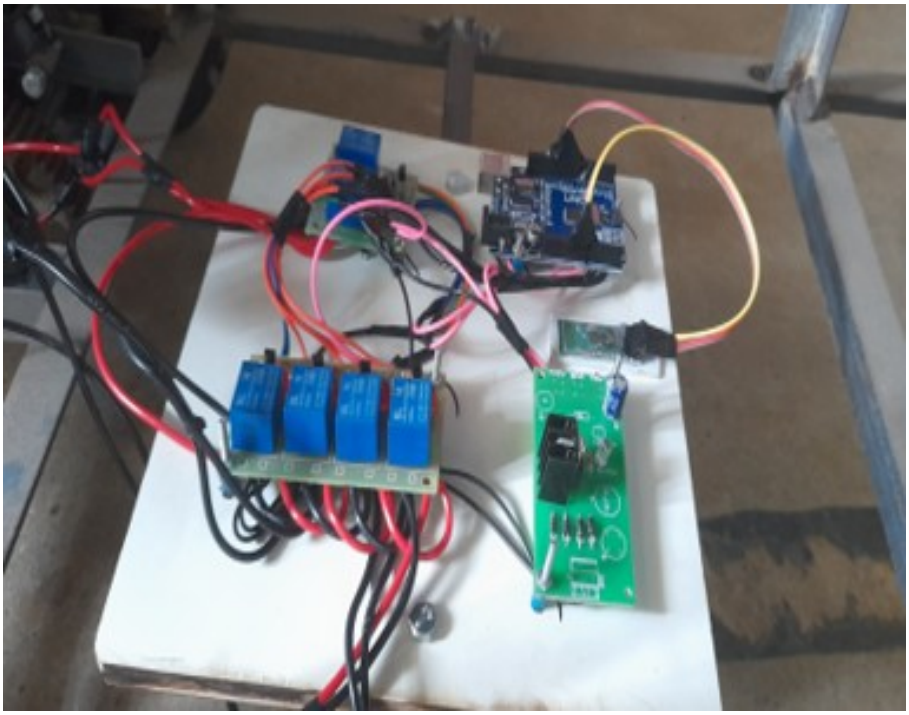


Figure 5.2: Circuit Board used for electronic integration

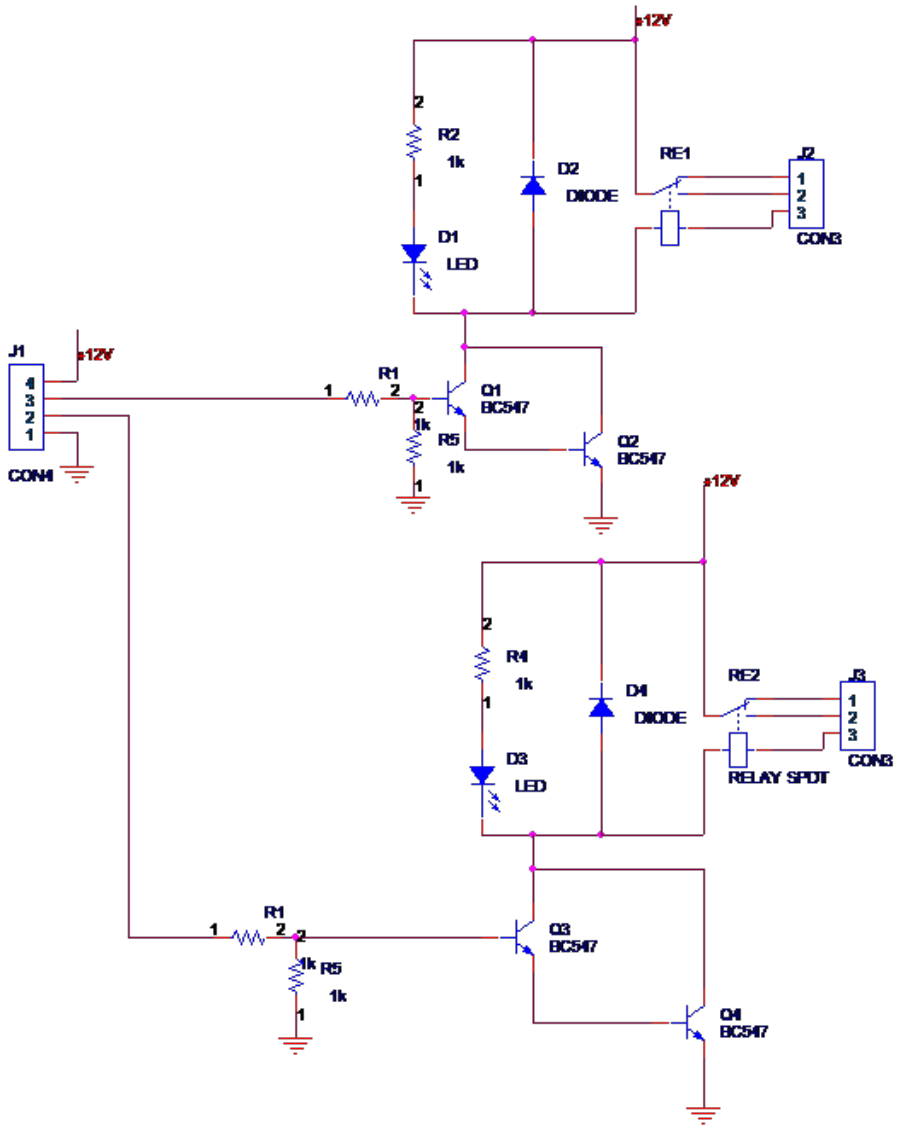


Figure 5.3: Schematic Circuit diagram

5.4 Final Assembly

The final assembly stage involved integrating the fabricated mechanical components with the electronic control system.

The base frame was first assembled and fitted with fiber wheels to allow easy mobility of the unit. The rack gear was securely mounted onto the sliding mechanism attached to the backrest, while the pinion gear was fixed to the motor shaft using a key connection.

Ball bearings (No. 6202) were installed in the bearing housings to support the shaft

and ensure smooth rotation with minimal friction. The electronic components, including the battery, motor driver, and control circuitry, were enclosed within a protective casing beneath the main frame to prevent damage and ensure safe operation.

After the assembly was completed, a trial run was conducted to verify the functioning of the system. The rotation of the pinion gear successfully produced the required linear displacement of the rack, enabling the mechanism to smoothly transition between bed and chair configurations.

The voice recognition module was then integrated with the Arduino control unit to enable hands-free operation of the mechanism. The module was programmed to recognize predefined commands such as “Bed” and “Chair,” which trigger the motor driver to rotate the motor in the required direction. This integration allows users with limited mobility to adjust the position of the chair–bed mechanism without physical effort.

To enhance safety and reliability, limit switches were installed at both ends of the rack travel path. These switches automatically stop the motor when the mechanism reaches its maximum allowable displacement, preventing mechanical overtravel and potential damage to the system. The successful integration of mechanical and electronic components ensured smooth operation, structural stability, and user-friendly control of the automated chair–cum–bed mechanism.



(a) Isometric view of the assembled chair cum bed prototype showing the structural integration of the main chassis and the motorized backrest.



(b) Side profile of the fabricated mechanism in the chair configuration, demonstrating the compact ergonomic design.



(c) Rear elevation of the final prototype, detailing the rack and pinion transmission system and motor mount configuration.

Figure 5.4: Multi perspective views of the fabricated automated chair cum bed prototype, illustrating the structural assembly and mechanical drive integration.

Chapter 6

TESTING AND RESULTS

6.1 Prototype Testing

The fabricated prototype was subjected to functional testing to evaluate the performance of the voice-controlled mechanism and the efficiency of the rack-and-pinion system.

The voice recognition module successfully detected the commands “Bed” and “Chair,” and the corresponding mechanical movement was initiated within approximately 1.5 seconds after the command was given. The transition between positions was observed to be smooth and stable.

6.2 Load Testing and Observations

Load testing was carried out by applying static loads to the chair–bed structure in order to evaluate its structural stability and mechanical performance.

Observations: The rack-and-pinion mechanism successfully withstood the calculated downward force without any signs of gear tooth slipping or shaft deformation. The system was also capable of handling additional load conditions during human weight simulation.

Torque Verification: The motor was capable of delivering the required torque of approximately 10 Nm, which was sufficient to overcome the frictional resistance

of the bearings and the weight of the moving frame. The motion of the mechanism was smooth and free from jerks or vibration.

6.3 Performance Evaluation

The experimental results confirmed that the actual rack length of 215 mm was sufficient to achieve the required range of motion, providing a safe margin above the calculated minimum travel distance of 194.76 mm.

Furthermore, the use of the worm gearbox effectively prevented back-driving of the motor under load conditions. This ensured that the mechanism remained stable and secure even when supporting patient weight, thereby meeting the safety requirements of the design.

Chapter 7

CONCLUSION

The voice-controlled chair–bed mechanism was successfully designed and developed with the objective of reducing human effort and improving convenience for patients with limited mobility. The system integrates a rack-and-pinion mechanism with an electronic control unit to enable smooth transformation between chair and bed configurations. The prototype operates using a 12 V battery-powered DC motor that provides consistent and reliable motion for the mechanism, while the worm gear arrangement ensures a self-locking feature that prevents unintended movement when the power supply is disconnected. This improves the safety and stability of the system during operation. The developed mechanism is particularly beneficial for physically challenged or bedridden individuals, as it reduces the need for external assistance and allows the user to control the movement using simple voice commands. Compared to conventional systems where another person is required to adjust the chair or bed position, the proposed system offers greater independence and convenience. In addition, the mechanical components used in the design require relatively low maintenance and provide reliable performance. Although the initial cost of the system may be slightly higher than traditional manual mechanisms, the advantages of automation, ease of operation, and improved comfort make it a practical assistive solution. Overall, the project demonstrates the successful integration of mechanical design and electronic control to develop an efficient and user-friendly assistive device that has potential applications in healthcare and rehabilitation environments.

Bibliography

[1] Tan Kian Hou, Yagasena, and Chelladurai, *Arduino based voice controlled wheelchair*, Journal of Physics: Conference Series, Vol. 1432, No. 012064, 2020.

[2] K. Velusamy, Kishore Kumar M., Naveen Kumar N., Nishanth E., and Ramanasarathi A. T., *Design and Analysis of Voice Control Wheel Chair*, International Journal of Engineering Science and Computing (IJESC), Vol. 9, Issue 3, 2019.

[3] D. Vijendra Babu, P. Subramanian, and R. Karthikeyan, *Voice Controlled Wheel Chair*, International Journal of Pure and Applied Mathematics, Special Issue, Vol. 119, No. 16, pp. 2157-2161, 2018.

[4] PSG College of Technology, *PSG Design Data: Data Book of Engineers*, Kalaikathir Achchagam, Coimbatore, ISBN: 978-8192735504.

[5] K. Mahadevan and K. Balaveera Reddy, *Design Data Handbook for Mechanical Engineers in SI and Metric Units*, CBS Publishers & Distributors Pvt. Ltd.

APPENDIX

- Program of Arduino UNO R3
- Power Point Presentation of the Project

Arduino Source Code

```
#include <SoftwareSerial.h>
#include <Wire.h>
#include <LiquidCrystalWired.h>

#define LCD_ADDRESS 0x3E
#define LCD_COLUMNS 16
#define LCD_ROWS    2

LiquidCrystalWired lcd = LiquidCrystalWired(LCD_ROWS, LCD_COLUMNS
    , FONT_SIZE_5x8, BITMODE_8_BIT);
SoftwareSerial Genotronex(10, 11); // RX - 11, TX - 10

int Rel1 = A0;
int Rel2 = A1;
int Rel3 = A2;
int Rel4 = A3;
int Rel5 = 4;
int Rel6 = 5;
int data;

void setup() {
  Serial.begin(9600);
  lcd.begin(LCD_ADDRESS, &Wire);
  lcd.turnOn();

  lcd.clear();
  lcd.setCursorPosition(0, 0);
  lcd.print("BLUETOOTH VOICE");
  lcd.setCursorPosition(1, 0);
  lcd.print("CONTR WHEELCHAIR");
  delay(2000);

  Genotronex.begin(9600);

  pinMode(A0, OUTPUT);
  pinMode(A1, OUTPUT);
```

```
  pinMode(A2, OUTPUT);
  pinMode(A3, OUTPUT);
  pinMode(4, OUTPUT);
  pinMode(5, OUTPUT);

  digitalWrite(A0, LOW);
  digitalWrite(A1, LOW);
  digitalWrite(A2, LOW);
  digitalWrite(A3, LOW);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
}

void loop() {
  if (Genotronex.available()) {
    data = Genotronex.read();
    Serial.print(data);

    if ((data == 'A')) {
      digitalWrite(Rel1, HIGH);
      digitalWrite(Rel3, HIGH);
      lcd.clear();
      lcd.setCursorPosition(0, 0);
      lcd.print("WHEEL CHAIR");
      lcd.setCursorPosition(1, 0);
      lcd.print("FORWARD MOVING");
    }

    if ((data == 'a')) {
      digitalWrite(Rel1, LOW);
      digitalWrite(Rel3, LOW);
      lcd.clear();
      lcd.setCursorPosition(0, 0);
      lcd.print("WHEEL CHAIR");
      lcd.setCursorPosition(1, 0);
      lcd.print("FORWARD STOP");
    }

    if ((data == 'E')) {
```

```
75        digitalWrite(Rel5, HIGH);
76        lcd.clear();
77        lcd.setCursorPosition(0, 0);
78        lcd.print("WHEEL CHAIR");
79        lcd.setCursorPosition(1, 0);
80        lcd.print("CONVERT INTO BED");
81        delay(3000);
82        digitalWrite(Rel5, LOW);
83      }
84
85      if ((data == 'G')) {
86        digitalWrite(Rel1, LOW);
87        digitalWrite(Rel2, LOW);
88        digitalWrite(Rel3, LOW);
89        digitalWrite(Rel4, LOW);
90        digitalWrite(Rel5, LOW);
91        digitalWrite(Rel6, LOW);
92        lcd.clear();
93        lcd.setCursorPosition(0, 0);
94        lcd.print("WHEEL CHAIR");
95        lcd.setCursorPosition(1, 0);
96        lcd.print("STOP");
97      }
98      delay(50);
99    }
100 }
```

Listing 7.1: Program of Arduino UNO R3

Design and Fabrication of an Automated Chair Cum Bed Mechanism for Bedridden Patients

Group number: A4

Group members

ABIN C S - VAS22ME006

BLESONJOSEPH – VAS22ME024

GODSON V O – VAS22ME026

GUIDE

Mr. VISHNU SHAJI

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Contents

- Introduction
- Objectives
- Problem Definition & Scope
- Methodology
- Design / Model Development
- Simulation / Experimental Setup
- Results & Discussion
- Validation
- Conclusion
- Future Work
- References
- Q&A

Introduction

- Hydraulic cylinders and pumps increase overall bed weight and make handling more complex.
- Hydraulic fluid systems are prone to leakage, affecting hygiene and patient safety.
- Hydraulic actuators require frequent maintenance and inspection to remain reliable.
- Manual or valve-based hydraulic controls offer less precision in motion and positioning.
- Mechanism Type: Rack and pinion system driven by an electric motor.
- Motion Conversion: Converts rotary motion into accurate linear displacement.
- Fluid-Free Operation: Eliminates the need for hydraulic fluids, reducing mess
- Compact Design: Enables sleeker, more space efficient bed structures.
- Low Maintenance: Mechanical components are easier to service and more durable.

Objectives

- To design and develop a rack and pinion mechanism that replaces hydraulic lifting in adjustable hospital beds and Home.
- To select appropriate motor, gearbox, and materials ensuring compactness and durability.
- To demonstrate a cost effective, maintenance free, and hygienic alternative to hydraulic systems.
- To ensure safe and precise operation of the chair cum bed system by using sensors and limit switches for motion control and hazard prevention.
- To Automation and multilanguage voice command.

Problem Definition & Scope

- **What**: Hydraulic actuation system are inefficient, heavy, and hard to maintain, lacking precise control for critical care settings.
- **Where**: The challenge is most pressing in hospitals and homes for immobile patients and individuals with limited mobility.
- **Why**: Manual or fluid-based actuation increases risk, effort, and cost, limiting patient independence and caregiver efficiency. Automated systems with rack-and-pinion and using single-language voice controls, offer safer, more energy-efficient, and accessible solutions for patient comfort and autonomy.

5

Scope and Limitations

Scope

- Design and develop a voice controlled chair cum bed system for patient comfort.
- Replace hydraulic actuation with rack and pinion mechanism .
- Improve hygiene, safety, and maintenance through electromechanical design.
- Enable smooth head and leg section movement using basic automation.

Limitations

- Work limited to conceptual and simulated models (no full prototype).
- Uses basic electromechanical controls; no advanced robotics or AI.
- Voice control supports only one language (e.g., English).
- Budget and material constraints restrict full-scale testing and fabrication.
- Preliminary design calculations may require revision for real-world deployment.

6

Literature Survey

YEAR & AUTHOR	METHODOLOGY	KEY FINDINGS	GAP
2020, Tan Kian Hou, Muhammad Khairi Bin Abd Amir, Siti Nurul Akmal Binti Yusof, and Rohana Binti Abdul Karim.	Utilized an Arduino Uno integrated with an HC-05 Bluetooth module. Voice commands are processed via an Android application (Google Voice) and sent to the microcontroller to drive DC motors through an L298N Motor Driver.	The prototype successfully responded to five basic commands (Forward, Backward, Left, Right, Stop). It proved to be a low-cost solution by retrofitting an existing manual wheelchair.	The system lacks an autonomous obstacle detection or avoidance system, making it potentially unsafe in crowded or unfamiliar environments.
2019, K. Velusamy, M. Kishore Kumar, N. Naveen Kumar, E. Nishanth, and A. T. Ramanasarathi.	Similar Arduino-Bluetooth architecture, but includes an **Ultrasonic Sensor** for safety. It uses an Android interface to manage graphical commands and speech recognition.	The inclusion of ultrasonic sensors allowed the wheelchair to detect hurdles, adding a layer of safety that basic models lack. It emphasizes the "daily need" aspect of independence.	The paper notes that environmental noise can interfere with speech recognition accuracy, and the system is limited by the range of the Bluetooth connection.

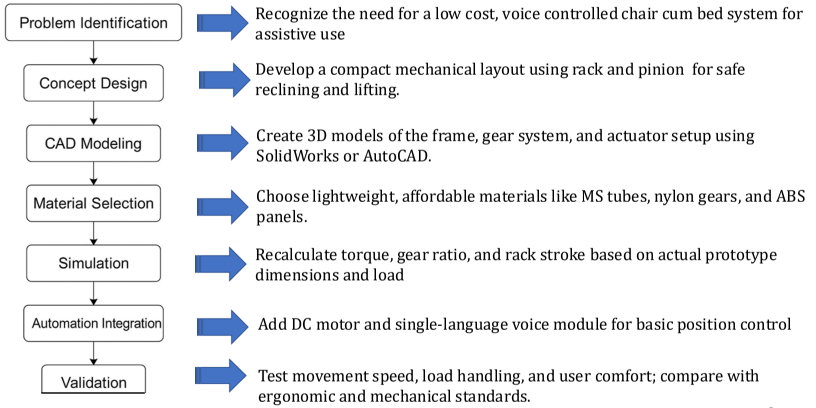
7

Literature Survey

YEAR & AUTHOR	METHODOLOGY	KEY FINDINGS	GAP
2018, D. Vijendra Babu, P. Subramanian, and R. Karthikeyan.	Employs Hidden Markov Models (HMM) for the speech recognition engine. The control unit is developed under the ARIA environment. It focuses on the signal processing aspect of voice commands.	The use of HMM improved the recognition accuracy of oral commands compared to simpler "plug-and-play" app modules.	The complexity of the ARIA environment and HMM might make the system harder to maintain or repair for a general user compared to standard Arduino setups.
2017, Mrs. M. D. Balsaraf, Mrs. V. S. Takate, and Mr. Bachkar Siddhant.	Unlike the others, this uses an IOIO Board (pronounced "yo-yo") rather than an Arduino. It focuses on a touch-based Android interface rather than just voice.	The IOIO board allowed for a more direct "plug-and-play" relationship with Android devices. The system proved very stable for manual directional control via a smartphone screen.	The restriction of the project was that the user can only touch one button at a time, and it lacks the hands-free benefit provided by voice-controlled models.

8

Methodology



System Setup

1. Description of Setup / Test Rig

- Test rig for voice controlled chair cum bed with head section movement.
- Mounted on rigid frame with rack and pinion mechanism .
- Purpose: verify linear displacement, torque transmission, and control accuracy under compact size and weight constraints.
- Sensors: rotary encoder (motor shaft), limit switches, current sensor for obstruction detection.
- Control: microcontroller based system with single language voice command (English).
- Safety: manual override, soft start or stop, and mechanical limiters to prevent overtravel

2. Software / Hardware Details

- Software: AUTOCAD & SolidWorks
- Hardware: DC motor, rack and pinion , limit switches ,sensors
- Control: Arduino with single-language voice command.
- Outputs: rack movement, motor current, torque , stroke time.

COMPONENTS

Table 1: Components List

Sl. No	NAME OF THE COMPONENT	MATERIAL	QUANTITY
01.	12V DC Motor	Aluminum	3
02.	12 V Battery 7 AH	Lead-Acid	1
03.	Shaft	M.S	2
04.	Chassis	M.S	1
05.	Chair arrangement	Pipe & Channel(M.S)	1
06.	Bolt, Nut and Washer	Steel	10
07.	Rack and Pinion Gear	M.S	1
08.	Wheel	Fiber	4
09.	Bearing	Steel	4
10.	Bearing Cap	M.S	4
11.	Spur gear	C.I	4



Fig 1: DC MOTOR



Fig 2: CASTER WHEEL



Fig 3: SPUR GEAR



Fig 4: WHEELS



Fig 5: LEAD ACID BATTERY



Fig 6: BALL BEARING

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Fig 7: ARDUINO UNO



Fig 8: HC-05 BLUETOOTH MODULE



Fig 9: POWER SUPPLY MODULE



Fig 10: SINGLE RELAY MODULE



Fig 11: 4 RELAY MODULE

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Design and Drawing

1. Load Analysis and Design Constants

Patient mass (m) = 100kg

Acceleration due to gravity (g) = 9.81m/s^2

Total Static weight(W) = $m \times g = 9.81\ N$

Factor of Safety(Fos) = 2.0 (to account for dynamic patient movement)

Design Load (F_d) = $W \times Fos = 981 \times 2.0 = 1962\ N$

Backrest Load (F_{back})= 58% of W = 0.58×981 =568.98

(Based on torso weight distribution) = 569N

Design and Drawing

2. Design of Pinion

Material = Cast Iron

Module(m) = 1.5mm

Number of Teeth(Z) = 21

Pitch circle Diameter (d)= $m \times z = 31.5$mm

Outer Diameter (d_o)= $d + 2m = 31.5 + 2 \times 1.5 = 34.5mm$

Total depth (h) = 2.25 × m = 2.25 × 1.5 = 34.5mm

Tangential Force(Ft) = Assuming two racks

$$Ft = F_{back}/2 = 284.5N$$

Design and Drawing

Lewis Strength Verification

$$W_t = \sigma_b \times b \times \pi \times m \times y$$

Allowable Stress (σ_b)= 60M Pa (Cast Iron)
Face Width (b) = 15mm($10 \times m = 10 \times 1.5$)
Form Factor (y) = 0.32 (For Z=21)
Wt = 60 × 15 × π × 1.5 × 0.32
= 60 × 15 × 4.71 × 0.32
= 900 × 1.5072 = 1356.48N

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Design and Drawing

Since the allowable strength 1356.48N is significantly higher than the actual applied load 284.5N, the cast iron pinion with 1.5 module and 21 teeth is safe for a100kg patient

As Calculated earlier,
Total force of patient = 100kg × 9.81 = 981N
Force on Backrest (58% of weight) = 569N
Force per Pinion (Ft)= 569/2 = 284.5N
The actual tangential for per pinion for a 100kg patient is 284.5N
Safety Factor check= 1356.48/284.5 = 4.7

Therefore the design is safe as the permissible load exceeds the actual load by a factor of 4.7

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Design and Drawing

3. Design of Rack

Module(m) = 1.5mm
Cross-section = 20 × 15mm
Circumference or Rotation= π × d = 97.38mm
Actual Length Provided= 215mm
Number of Teeth (Z)= 45

Design and Drawing

4. Design of Shaft and Bearing

Patient mass(m)=100kg
Design Load (w)= m × g = 100 × 9.81 = 981N
Load Per Shaft = Assuming the mechanism uses two shafts or a split load, F=490.5N
Motor Power(P)= 2HP ≈ 1.5Kw
Motor Speed(N)= 1440 rpm
Material Allowable Shear stress(τ)= 40MPa ($40N/mm^2$)

Design and Drawing

(i) Calculation of Tortional Moment(M_t)

$$T = \frac{60 \times P}{2\pi \times N}$$

= 9.947 ≈ 10Nm

Note; (in Nmm for stress calculations: 10,000Nmm)

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Design and Drawing

(ii) Calculation of Bending Moment(M_b)

Mb = F × L

Mb = 490.5 × 40mm

= 19620N mm

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Design and Drawing

(iii) Equivalent Twisting moment (M_e)

$$Me = \sqrt{(Mb)2 + (T)2}$$

$$= \sqrt{(19620)^2 + (10,000)^2}$$

$$= \sqrt{384,944,400 + 100,000,000}$$

$$= \sqrt{484,944,400}$$

$$= 22,021.5\,Nmm$$

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Design and Drawing

5. Determination of Minimum Shaft Diameter(d)

$$\tau = \frac{16 \times M_e}{\pi \times d^3}$$

$$d = \sqrt[3]{\frac{16 \times Me}{\pi \times \tau}}$$

$$= \sqrt[3]{\frac{352344}{125.66}} = \sqrt[3]{\frac{16 \times 22021.5}{\pi \times 40}}$$

$$= \sqrt[3]{2803.9} = 14.1mm$$

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Design and Drawing

Therefore, Calculated Minimum Diameter = 14.1mm

Actual Shaft Diameter Selected = 15mm

Since the actual diameter is greater the calculated minimum required diameter(14.1mm),

The shaft is safe and will not undergo permanent deformation or failure under the weight of a 100kg patient.

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SPECIFICATION OF PINION

Material: Cast Iron

Outer diameter(do): 34mm	do=di+2a (Reference from Web)
Circular pitch(p): 4.7mm	(From Design Data book by Mahadevan, Pg: 203, Eqn 12.1(d))
Total depth(h): 3.375mm	(Addendum + Dedendum)
Module(m): 1.5mm	(From Design Data book by Mahadevan, Pg: 203)
Pressure angle: 21°	
Pitch circle diameter(d): 31mm	(From Design Data book by Mahadevan, Pg: 203, Eqn 12.1(e))
Addendum(a): 1.5mm equal to module	(Reference from the Web)
Dedendum(hf) 1.875mm standard	
Circular tooth thickness(t): 2.355mm	Circular pitch (p) divided by 2
Fillet radius(r): 0.45mm	Standard= 0.3m to 0.4m
Clearance(c): 0.375mm	Dedendum- addendum
Number of Teeth(z)= 21	

SPECIFICATION OF RACK

Material : Cast iron	
Pitch Diameter(d): 97.38	
Calculated length of the rack: 194.76mm	
Actual length taken: 215mm	
Module: 1.5mm	(From P.S.G Design Data book, Pg: 203)
Cross-section: 20 x 15mm	
Number of teeth (Z): 45	(Length of rack divided by pitch diameter of rack)

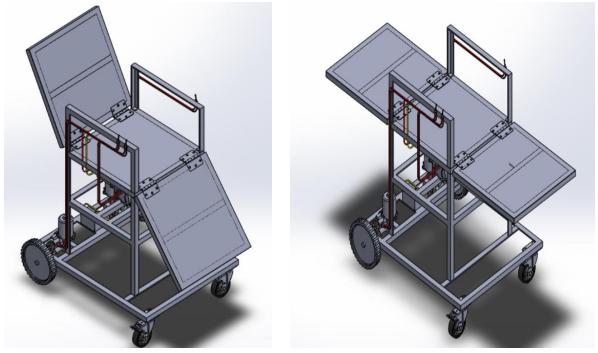


Fig 12:Front view

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Fig 14: Back view

Fig 13: Top view

Fig 15: Side view

Fig 16: WHEEL CHAIR BED LAYOUT SHOWING THE RACK PINION MECHANISM BELOW THE CHAIR

WHEEL CHAIR CUM BED LAYOUT

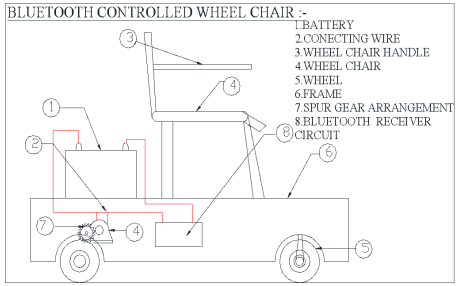


Fig 17: WHEEL CHAIR BED LAYOUT SHOWING THE SHAFT AND GEAR MECHANISM ON THE REAR WHEELS

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RACK GEAR

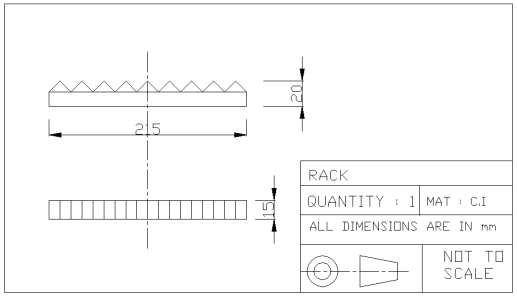


Fig 18: RACK LAYOUT

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PINION GEAR

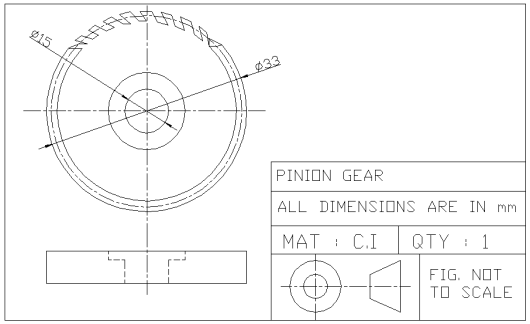


Fig 19: PINION LAYOUT

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SHAFT

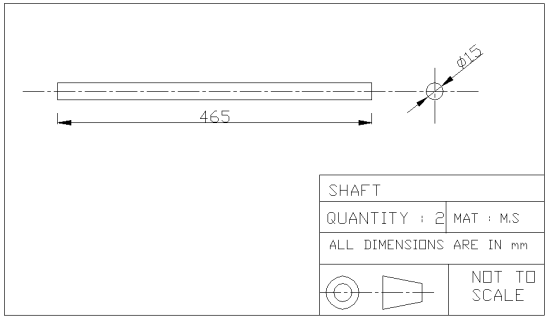


Fig 20: SHAFT LAYOUT

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BALL BEARING-6202

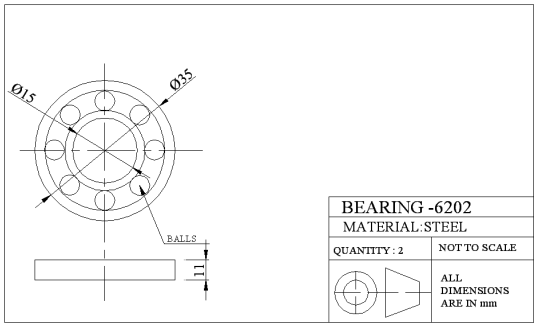


Fig 21: BALL BEARING LAYOUT

Fabrication and Assembly Process



Fig 24: Photos showing team members working on Fabrication and Assembly processes

WORKING PRINCIPLE

Power Supply and Control System

- A **Lead-Acid Battery** is used as the main power source for the project.
- The output from the lead-acid battery is supplied to the **Bluetooth receiver circuit**.
- The Bluetooth receiver module controls the movement of the system wirelessly.
- The receiver unit consists of **four relays**.
- These four relays are connected to **two DC motors** for directional control.

Relay Operation for Direction Control

- **Relay 1 & Relay 3** → Forward Direction
- **Relay 2 & Relay 4** → Reverse Direction
- **Relay 1 & Relay 4** → Left Direction
- **Relay 2 & Relay 3** → Right Direction

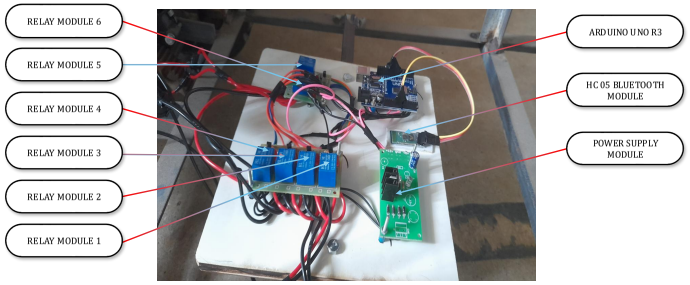


Fig 22: CIRCUIT BOARD

System Operation and Functional Testing

- Trial run was conducted after assembly to verify proper functioning of the mechanism.
- Rotation of the pinion gear converts rotary motion into linear displacement of the rack.
- This movement enables smooth transition between bed and chair configurations.
- A voice recognition module was integrated with the Arduino control unit.
- The system recognizes commands such as ***'Forward', 'Backward', 'Left', 'Right', 'Bed', 'Chair' and 'Stop'***
- Voice commands control the motor driver, allowing hands-free operation for users with limited mobility.

Final Assembled Prototype



Fig 25: The images shows the Front view, Side view and Back view of fabricated voice-controlled chair-cum-bed prototype.

GANTT CHART

Table 2: GANTT CHART

WORKS	July	August	September	October	November	December	January	February	March
Group Formation									
Topic selection									
Literature Survey									
Concept design									
Modelling using software									
2D Drawing & Designing									
Cost Estimation & Calculation									
Purchasing									
Fabrication									
Testing									
Report Preparation									
			COMPLETED						
			ONGOING						
			UPCOMING						

Bill of Materials (BOM) and Cost Analysis

Table 3: Bill of materials

SL.NO	NAME OF THE PARTS	MATERIAL	QUANTITY	AMOUNT (RS)
1	Microprocessor & Bluetooth Control unit	HC-05 & Arduino UNO Rev3 Unit	1	3,000
2	D.C. Motor (12 V)	Aluminium	2	1,250
3	Bearing with bearing cap	Steel	4	1,000
4	Relay Driver unit	Electronic	1	350
5	Battery	Lead-acid	1	1,200
6	Spur Gear	M.S	2	1,200
7	Rack & Pinion Mechanism	M.S	1	850
8	Connecting Wire	Cu	2 meters	200
9	Bolt & Nut	M.S	12	400
10	Shaft	M.S	2	1,000
11	Frame Stand and sheet Metal	M.S	1	4,500
12	Wheel	Rubber	4	800
13	Labour Cost	LATHE, DRILLING, WELDING, GRINDING, POWER HACKSAW, GAS CUTTING:		2,500
14	Transportation			2,000
			TOTAL COST	20,000

Conclusion

- Successfully designed and built Wheelchair-cum-bed system with smooth motion using rack-and-pinion.
- Ensured safe and slow movement suitable for immobile patients.
- Integrated simple voice control for basic position commands.
- Achieved a compact, low-cost solution with potential for healthcare use.
- Offers hygienic, user-friendly operation and scope for further improvement.

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Q&A

- Thank you for your attention!
- We welcome your questions.

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